Basic Principles of Medicine 1
Module : Foundations to Medicine
DLA

Early Embryology Week 2

Recommended Reading

Moore KL, Persaud TVN, Torchia MG. The Developing Human – Clinically Oriented Embryology, 11th Ed. Chapters 1-4. pgs. 37-39, 47 - 56

- Pay special attention to the Green boxes for the clinical correlations for this topic.

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1022	Describe the differentiation of the trophoblast, formation of lacunar networks and the establishment of a primordial uteroplacental circulation.
SOM.MKII.BPM1.1.FTM.1.ANAT.1023	Describe the sequence of events that leads to development of the amniotic cavity, bilaminar embryo, primitive umbilical vesicle (yolk sac), extraembryonic mesoderm and its clinical significance.

Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

The Big picture

Sperm → Zygote
Secondary oocyte → Zygote
Zygote → Morula → Blastocyst
Blastocyst → Outer Cell Mass → Trophoblast
Trophoblast → Syncytiotrophoblast; Cytotrophoblast
Blastocyst → Inner Cell Mass → Embryoblast
Embryoblast → Hypoblast → Umbilical vesicle
Embryoblast → Epiblast

Epiblast → Endoderm; Mesoderm; Ectoderm

Endoderm

- Epithelial lining of Gut and associated glands (liver, pancreas, gall bladder)
- Epithelial lining of the respiratory organs (trachea, bronchus, lungs), tonsils, middle ear and pharyngotympanic tube.
- Parathyroid and thyroid glands
- Epithelial lining of bladder, most of urethra and lower vaginal canal

Mesoderm

Lateral

- Heart
- Spleen
- Adrenals
- Parietal / Somatic Layer
- Visceral / Splanchnic Layer

Intermediate

- Urogenital organs and ducts

Paraxial

- Sclerotome - bone
- Myotome - muscle
- Dermatome - skin

Ectoderm

Ectoderm →
- Epidermis, nails, sweat glands, lens, anterior pituitary
- Sensory cells of the ear and olfactory epithelium, lower anal canal

Ectoderm → **Neuroectoderm** → Neural Tube; Neural Crest Cells

Neural Tube

- CNS
- Retina
- Pineal body
- Posterior pituitary

Neural Crest Cells

- PNS
- Melanocytes
- Glial & Schwann cells
- Cranial nerves
- Odontoblasts
- Connective tissue & bones of face & skull
- Cells of the adrenal medulla
- Bulbar and truncal ridges of the heart

The journey

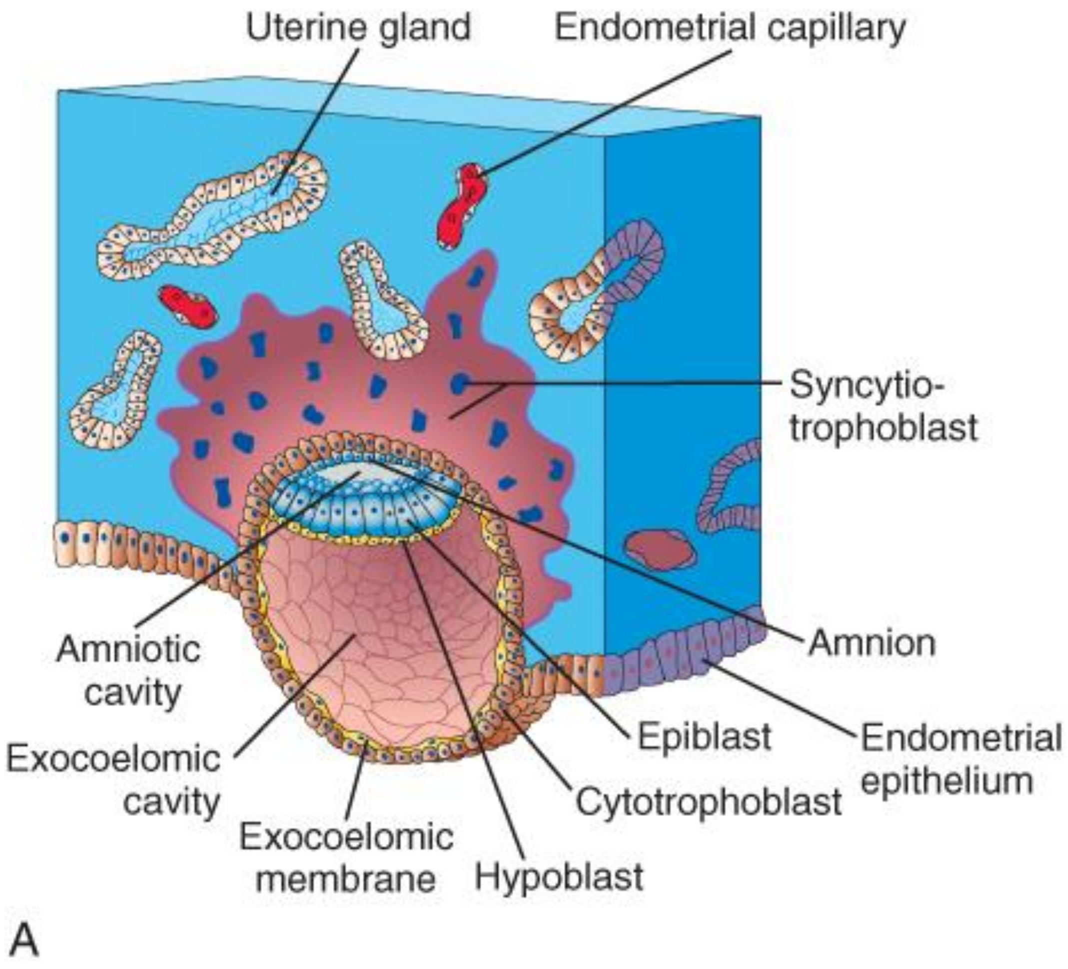
- The development of the embryo from male and female gametes to a fully formed embryo is termed Embryogenesis
- It consists of a series of events
  - Fertilization – Day 1
  - Cleavage – Day 1 - 6
  - Implantation – Day 6 – 10 (beginning of week 2)
  - Formation of the bilaminar disc – Day 10 - 14
  - Gastrulation (formation of embryonic cell lines) - Day 14 – 21 **(Week 3)**
- Organogenesis – **Week 4 to 8**
- Everything happens synchronously with many simultaneous events

Week 1

Week 2

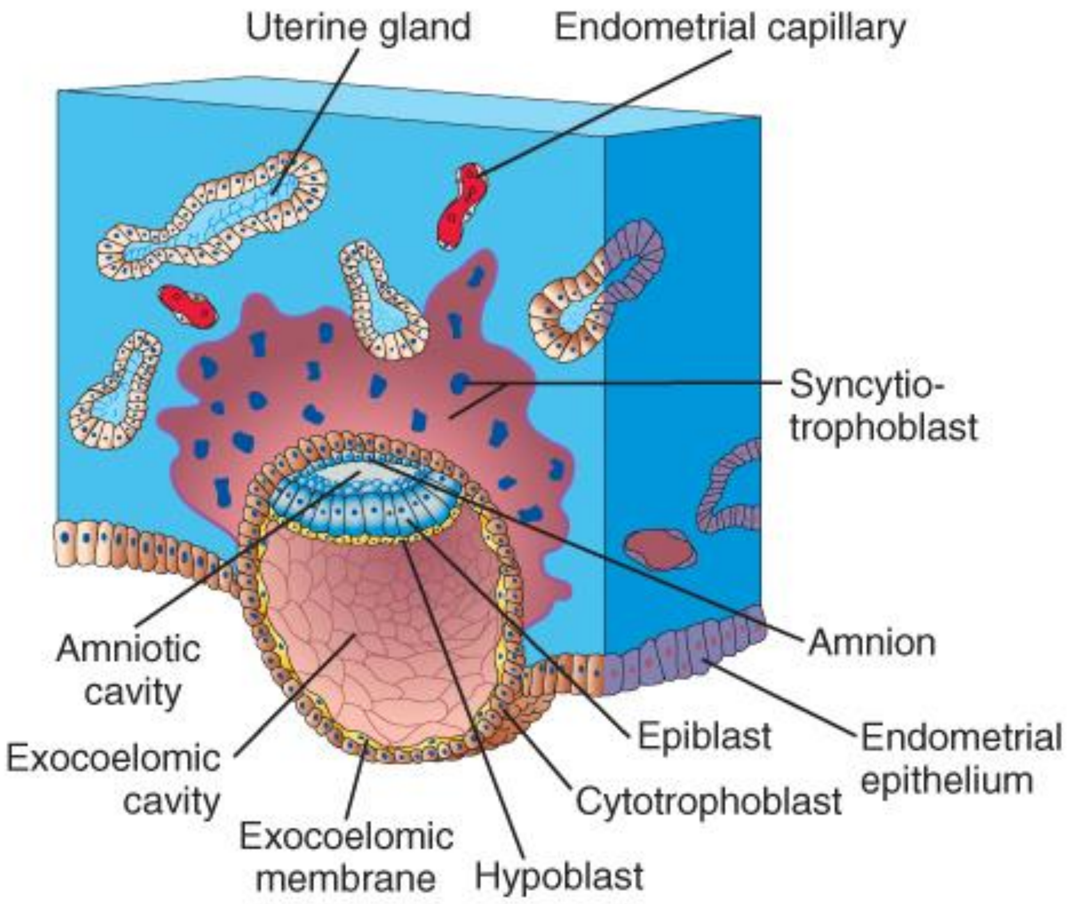
Week 2, Day 8 – Formation of the bilaminar disc

- Small space appears in the embryoblast = **amniotic cavity**
- Embryoblast forms a flat plate with two layers - **embryonic disc**
  - **Epiblast**, **high columnar** cells
    - Forms the floor of the amniotic cavity
  - **Hypoblast**, small **cuboidal** cells
    - Forms the roof of the exocoelomic cavity
    - Is continuous with the exocoelomic membrane



Week 2, Day 8 – Formation of the bilaminar disc

- **Amnioblasts**, from the epiblast form the amnion (lines the epiblast and the inner surface of cytotrophoblast)
  - secrete **amniotic fluid**
- The exocolomic cavity is now referred to as the **primary umbilical vesicle**



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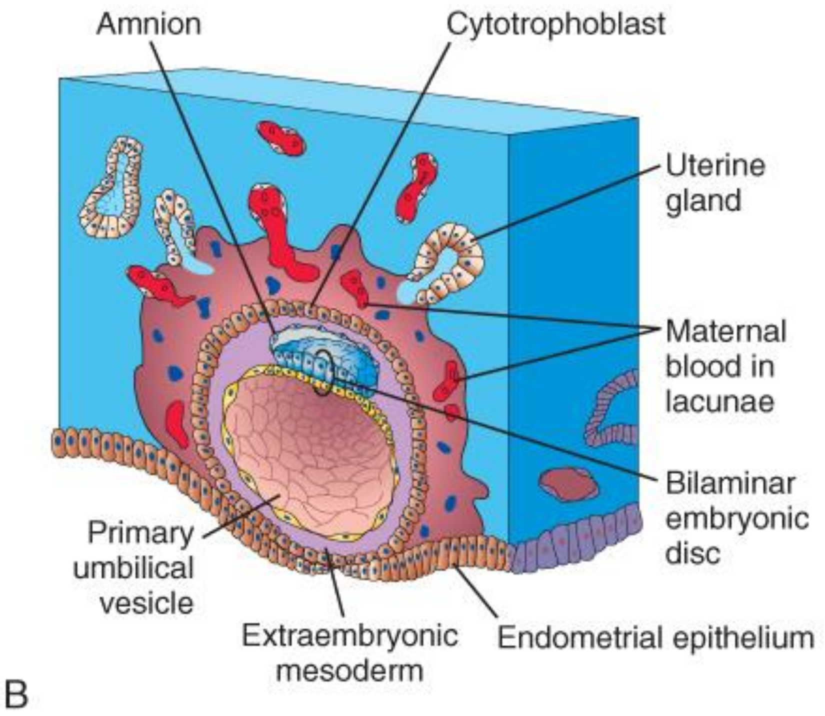
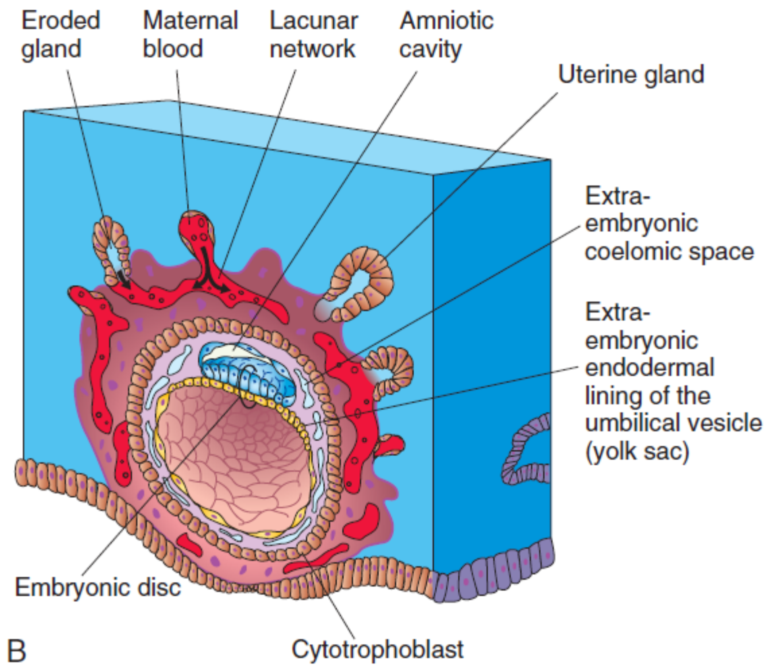
Week 2, Day 10 – 12, – Formation of the bilaminar disc

- Cells migrate from hypoblast to lie between the primitive umbilical vesicle and the trophoblast
  - Connective tissue called the **extraembryonic mesoderm**
  - Surrounds the amnion and umbilical vesicle.

- Spaces appear in the extraembryonic mesoderm
  - Rapidly enlarges to form the **extraembryonic coelom**

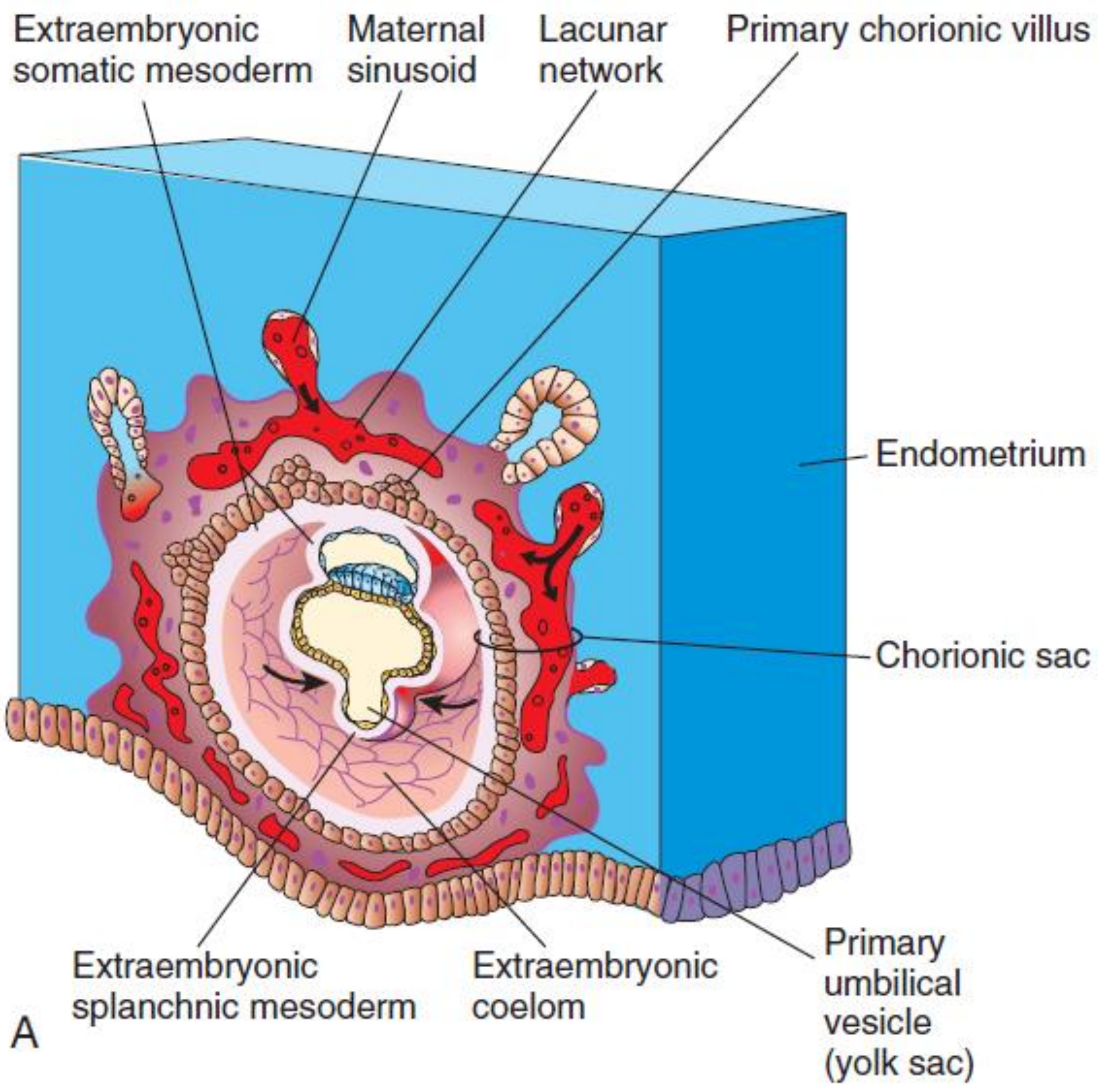
- Spaces appear in the syncytiotrophoblast = **Lacunae**
  - Filled with maternal blood
  - Forms **primordial uteroplacental circulation**





Week 2, Day 13 – Formation of the bilaminar disc

- The extraembryonic coelom divides the extraembryonic mesoderm:
  - **Extraembryonic somatic mesoderm Extraembryonic splanchnic mesoderm**
- Formation of **Chorion**
  - Extraembryonic somatic mesoderm
  - Syncytiotrophoblast
  - Cytotrophoblast
- The extraembryonic coelom eventually becomes the **chorionic cavity**

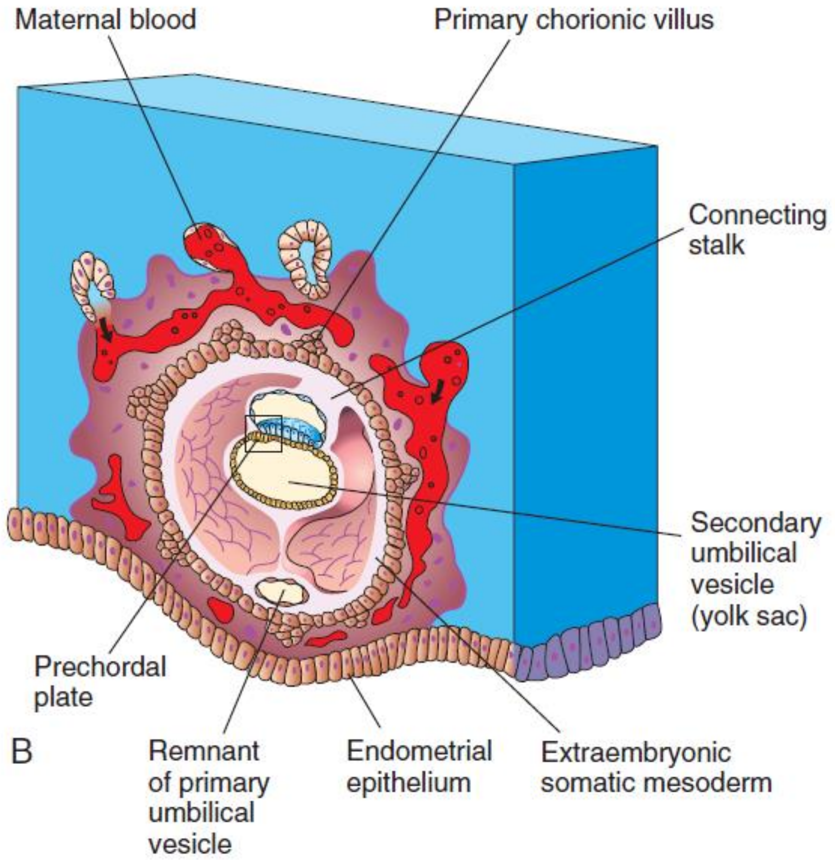


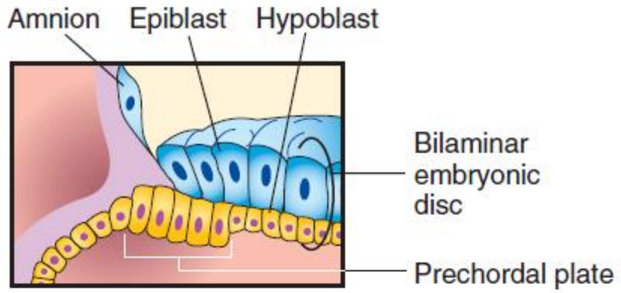
Week 2, Day 14 – Formation of the bilaminar disc

- The amnion and the primitive umbilical vesicle are suspended in the chorionic cavity by the **connecting stalk**

- **Extraembryonic endoderm cells migrate from the hypoblast and pinch of a section of the primary umbilical vesicle**
  - The chorionic cavity increases in size
  - Primary umbilical vesicle decreases in size and regress
  - Newly formed smaller **secondary umbilical vesicle**

- Hypoblastic cells at the cranial end of the embryo, change into columnar cells forming the **prechordal plate**

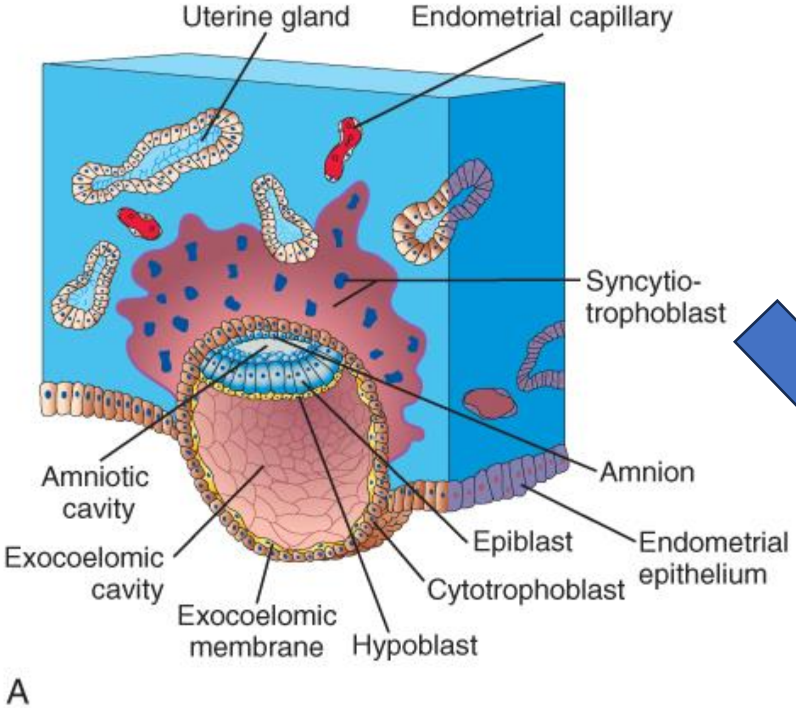


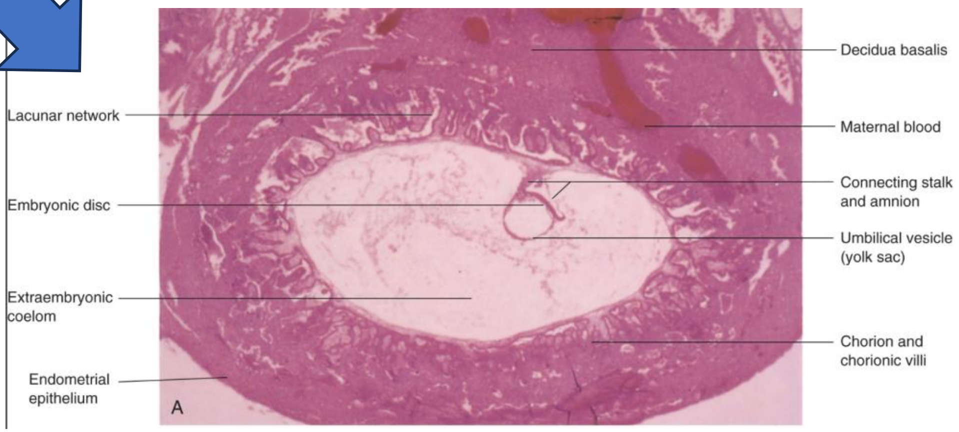


Summary of week 2

Sperm → Zygote
Secondary oocyte → Zygote
Zygote → Morula → Blastocyst → Outer Cell Mass → Trophoblast
Trophoblast → Syncytiotrophoblast
Trophoblast → Cytotrophoblast
Blastocyst → Inner Cell Mass → Embryoblast
Umbilical vesicle ← Hypoblast ← Embryoblast

- Amniotic cavity is formed
- Chorion and chorionic cavity is formed
- Primary umbilical vesicle is formed (and regresses)
- Secondary umbilical vesicle develops
- Bilaminar embryonic disc develops
- Prechordal plate develops





Formation of blood islands and establishment of primitive placenta

- Toward the end of week two cells within the extraembryonic mesoderm, connecting stalk and chorion start to differentiate into cardiogenic cells in patches called blood islands
- Embryonic blood vessels start forming 2 days later